# VAIBHAV SATISH DAHATONDE

**B.Tech Mechanical Engineering • Design Engineering & 3D Modelling**

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## Skills

**CAD & 3D Modelling:** **CATIA V5** (Preferred) • SolidWorks • Creo • Siemens NX • Fusion 360 • Sheet Metal & Assembly • Blender • 3DS Max

**Simulation & CAE:** ANSYS Mechanical • ANSYS CFX • Structural Analysis • Modal & Vibrational Analysis • CFD

**Design Competencies:** GD&T • Sheet Metal Design • Topology Optimisation • DFMEA • DFM • Engineering BOM

**Manufacturing:** CNC Operations • Precision Quality Inspection • Production Process Control • ESW

**Tools & Programming:** MS Excel • Python • ENOVIA / PLM Platforms • APQP (Awareness)

## Education

**B.Tech Mechanical Engineering** **: 8.04**
Vishwakarma Institute of Technology, Pune
2023 – 2027

**HSC** **81.83%**
Residential Junior College, Ahmednagar
2021 – 2023

**SSC** **89.20%**
Jawahar Madhyamik Vidyalaya, Chanda
2020 – 2021

## Certifications

- SolidWorks (CSWA) — Certified Associate
- ANSYS Mechanical — FEA & Simulation
- CATIA V5 — Part & Assembly Design
- NSS Best Volunteer — Camp 2024

## Leadership & Activities

### NSS Unit Coordinator
**Vishwakarma Institute of Technology** | 2023 – Ongoing
- **Led 50+ volunteers** for village infrastructure projects; managed scheduling, delegation, and government liaison.
- **Designed** official NSS unit branding end-to-end.
- Organised **NSS Camp 2024** — logistics for 100+ participants across 7 days.

### Chassis Domain Member
**Veloce Racing — SAE Club, VIT Pune** | 2023 – 2024
- Worked on **chassis design and structural analysis** of a Formula-style SAE race car — frame geometry, load path optimisation, and torsional rigidity using Solidworks, CATIA V5 and FEA.

## Summary

Final-year B.Tech Mechanical Engineering student at VIT Pune (**CGPA: 8.04**) with hands-on experience in 3D product design, structural simulation, and CAD modelling. Proficient in **CATIA V5, SolidWorks, Creo, and ANSYS** with completed engineering design projects covering structural optimisation, gearbox design, vibrational analysis, and CFD. Two CNC internships applying GD&T, DFMEA, and quality inspection in live production environments.

## Experience

### Intern — Quality & Manufacturing Process
**Associate Circuits** | Ghaziabad | May 2025 – Jul 2025 (3 months)
- Applied **GD&T and Engineering Standard Work (ESW)** to monitor CNC production processes and ensure dimensional accuracy of high-precision mechanical components.
- Conducted **DFMEA and 7-Step problem-solving** to diagnose and resolve recurring mechanical failures, reducing production rework.
- Managed high-accuracy CNC operations maintaining **100% adherence** to quality and delivery standards.

### Intern — Machine Integration & Commissioning
**NULED Electronics Pvt. Ltd** | Faridabad | Dec 2025 – Jan 2026 (2 months)
- Managed **mechanical and electrical commissioning** of the GIGA 8888 CNC system, validating all mechatronic parameters to operational standards.
- Performed **precision spindle fault analysis and repair**, resolving alignment deviations through systematic technical troubleshooting.
- Configured **VFD controls and Safety Function Units (SFU)**, optimising machine performance with full industrial safety compliance.

## Projects

### Topology Optimisation — Washing Machine Sheet Metal Chassis
| SolidWorks + ANSYS | Jul – Dec 2025
- Achieved **18% weight reduction** while maintaining structural rigidity and vibration stability through iterative FEA-driven geometry refinement; applied **DFM principles** (bend radii, flange tolerances, weld locations) to ensure full producibility.

### Vibrational Analysis & Durability Testing — Vehicle Projector Lamp
| Industry-Sponsored, Whitelight lamp pvt ltd.| ANSYS Mechanical | Jul – Dec 2025
- Performed **modal and spectral analysis**; analysed stress amplification **(1g–30g)**, identified **3 critical natural frequencies**, and proposed geometry modifications shifting resonance bands outside the vehicle's operational excitation range.

### Design & Analysis — Two-Stage Helical Gearbox for Chain Conveyors
| Solidworks + ANSYS | Jan – May 2026
- Designed a **7.5 kW** industrial gearbox achieving **24:1 speed reduction** (1440 to 60 RPM); validated gear geometry, shaft sizing, and bearing selection analytically. Produced full **Engineering BOM and Solidworks assembly drawings** with GD&T callouts.

### CFD Analysis — Axial Compressor Rotor
| ANSYS CFX | Jan – May 2026
- 3D steady-state CFD at 16,043 RPM validating **1.58 pressure ratio** and **88.8% isentropic efficiency**; full pressure/velocity distribution mapping across rotor span.

### AI-Powered PLM Data Extraction Platform
| Industry-Sponsored, Atlas Copco Pune R&D | Jan – May 2026
- Built an **ENOVIA-integrated prototype** combining Be2Net and drawing databases; cut part-spec lookup from **17 min to under 10 sec** at **93% accuracy**. Deployed AI layer auto-classifying **material specs, tolerances, and revision history** from legacy drawings.